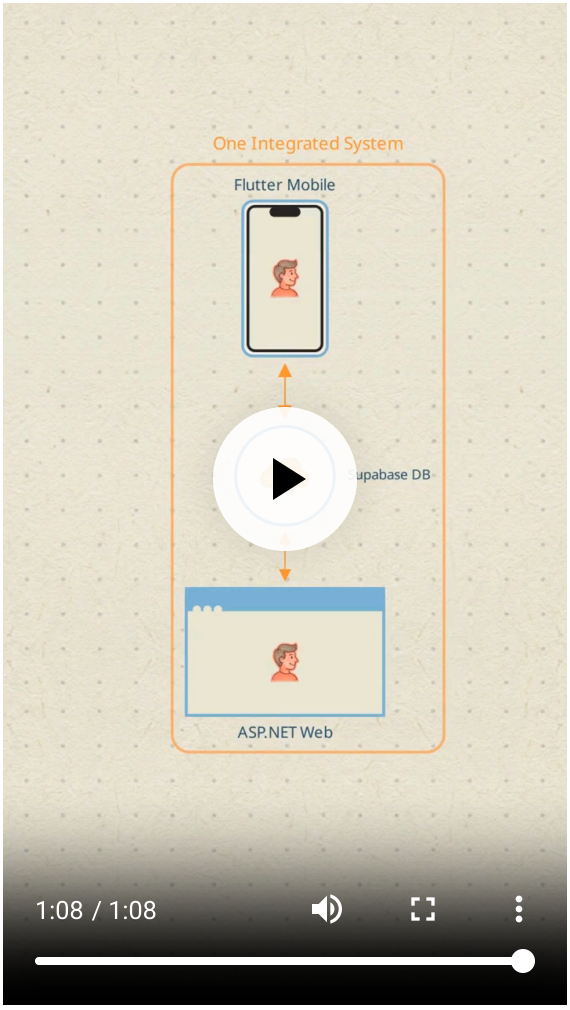
Unit 4 Overview

Welcome to Unit 4

In Unit 3, your group designed the system architecture, UML diagrams, database structure and user interfaces for the proposed solution.

In this unit, you will use those designs to develop the **Flutter mobile application and ASP.NET web application using the shared Supabase backend**. You will implement the approved functions, connect the applications to the backend and work towards a functioning integrated prototype.

The work completed in this unit will form the development and implementation component of **Assessment 3**.



Learning Outcomes

By the end of Unit 4, you should be able to:

1. Prepare the development environment for the proposed system.
2. Set up and configure the shared Supabase project.
3. Implement the database structure based on the approved ERD.
4. Configure authentication and appropriate user access where required.
5. Develop the main Flutter mobile application functions.
6. Develop the main ASP.NET web application functions.
7. Implement the approved user interfaces and navigation.
8. Connect both applications to the shared Supabase backend.
9. Implement the required database operations.
10. Demonstrate how information is shared between the mobile and web applications.
11. Use GitHub to manage source code and record development contributions.
12. Identify and record development challenges.
13. Update Microsoft Project and the Project Progress Tracker.
14. Prepare the integrated prototype for formal testing in Unit 5.

Unit 4 at a Glance

Activity	**What you will do**	**Main outcome**
Activity 4.1	Set up Supabase and implement the approved database structure	Shared backend
Activity 4.2	Develop the Flutter mobile application	Working mobile functions
Activity 4.3	Develop the ASP.NET web application	Working web functions
Activity 4.4	Connect the applications and implement shared processes	Integrated prototype

Documents and Tools Used in This Unit

Document or Tool	**How it will be used**
Updated SRS	Refer to the approved requirements and design artefacts
Flutter	Develop the mobile application
ASP.NET	Develop the web application

Supabase	Provide the shared backend, database and relevant services
GitHub	Manage source code, changes and development contributions
Microsoft Project	Track development tasks, responsibilities and progress
Project Progress Tracker	Record actual progress and individual contributions

How to Work Through This Unit

For each activity:

1. Watch, listen to or read the learning material.
2. Study the examples.
3. Complete the guided development task.
4. Apply the task to your own stakeholder project.
5. Check that the developed function works before continuing.
6. Commit your work to GitHub and keep the required evidence.
7. Update Microsoft Project and the Project Progress Tracker.
8. Complete the “Before Moving On” checklist.

Important

Continue working with the **same SRS, GitHub repository, Microsoft Project file and Project Progress Tracker** used in the previous phases.

Development must follow the approved requirements and system design. If an important change becomes necessary, record the reason and update the relevant documentation.

Both applications must use the **same Supabase backend** and form one integrated system.

By the end of Unit 4, your group should have an integrated prototype ready for the formal manual testing and corrections covered in Unit 5.

The completed work from **Unit 4 will form Assessment 3: System Development, Implementation, and Testing (50%)**.